

Microsserviços e DevOps com Spring Boot e Spring Cloud

DR3 — Assessment

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Relatório de evidências dos exercícios 1 a 12

Exercício 1 — produtos-service no Eureka

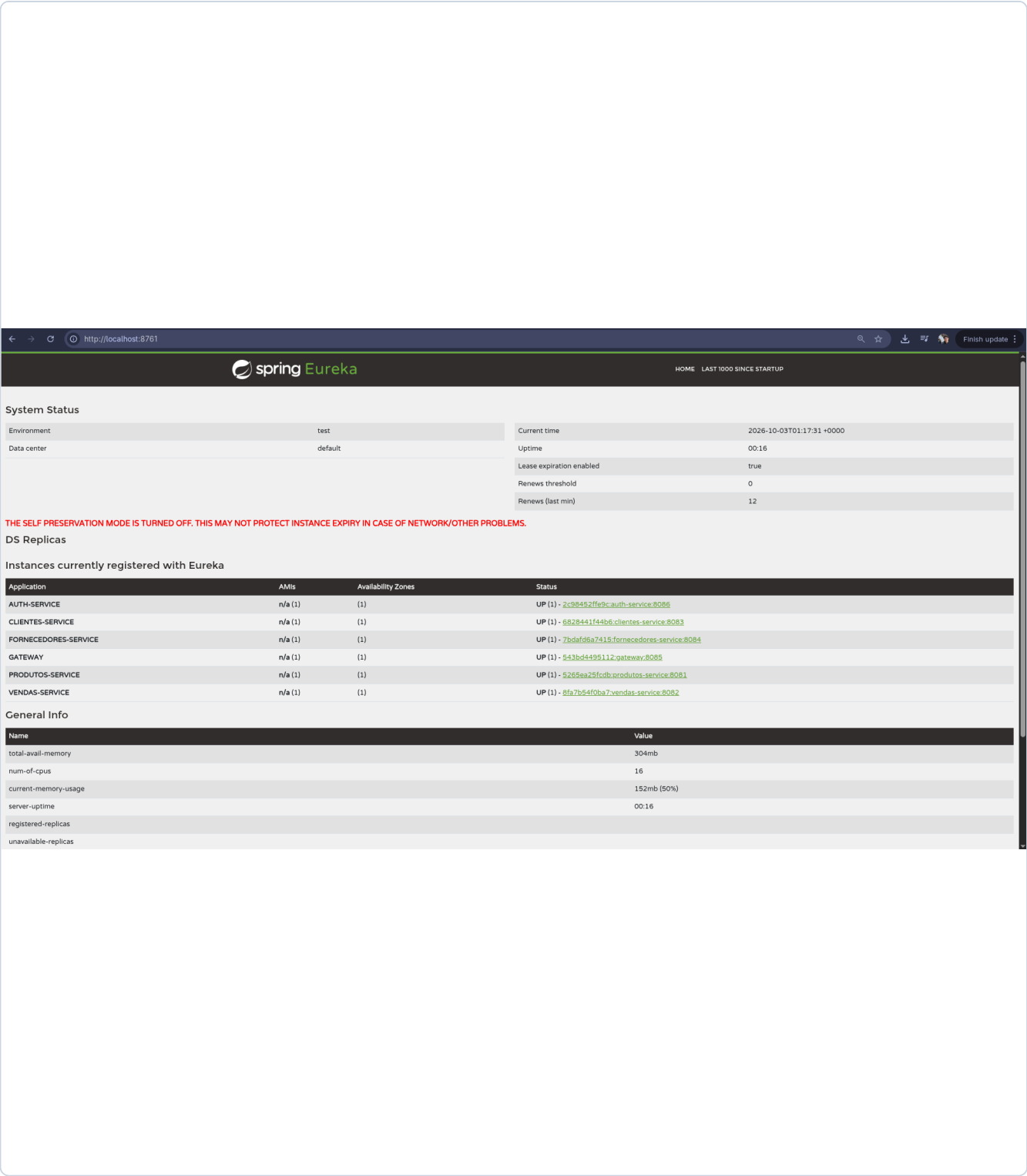


Figura 1 — Registro do produtos-service no Eureka.

Exercício 2 — Branch e Pull Request

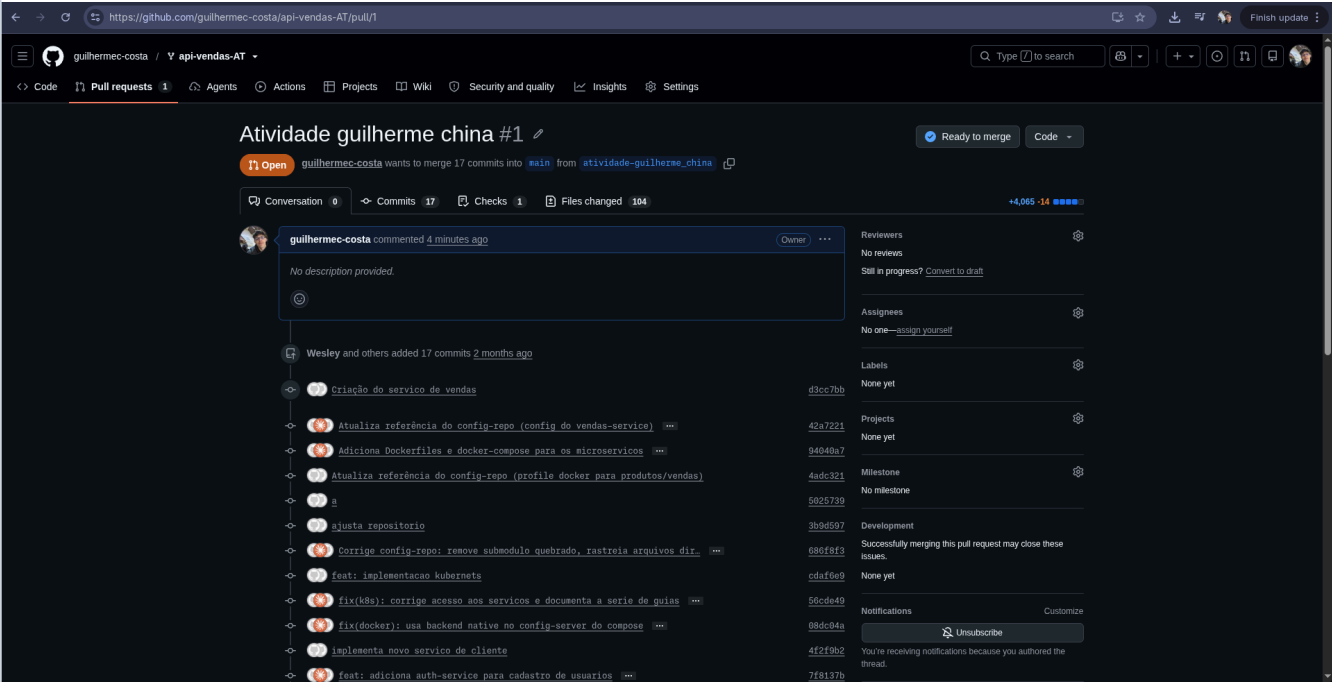


Figura 2 — Pull Request da atividade.

Exercício 3 — Inicialização na porta 8084

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
forneceadores-service | Hibernate: alter table if exists fornecedor drop constraint if exists UK_rhijyxxhmgutledr1hd15gp18
forneceadores-service | Hibernate: alter table if exists fornecedor add constraint UK_rhijyxxhmgutledr1hd15gp18 unique (cnpj)
forneceadores-service | 2026-10-03T01:01:40.880Z INFO 1 --- [forneceadores-service] [ main] j.LocalContainerEntityManagerFactoryBean : Initialized JPA EntityManagerFactory for persistence unit 'default'
forneceadores-service | 2026-10-03T01:01:41.066Z WARN 1 --- [forneceadores-service] [ main] JpaBaseConfiguration$JpaWebConfiguration : spring.jpa.open-in-view is enabled by default. Therefore, database queries may be
forneceadores-service | 2026-10-03T01:01:41.124Z INFO 1 --- [forneceadores-service] [ main] o.s.c.openfeign.FeignClientFactoryBean : For 'produtos-service' URL not provided. Will try picking an instance via load-bal
ancng.
forneceadores-service | 2026-10-03T01:01:41.426Z INFO 1 --- [forneceadores-service] [ main] DiscoveryClientOptionalArgsConfiguration : Eureka HTTP Client uses RestTemplate.
forneceadores-service | 2026-10-03T01:01:41.447Z WARN 1 --- [forneceadores-service] [ main] IgnitionLoadBalancerCaffeineWarnLogger : Spring Cloud LoadBalancer is currently working with the default cache. While this
cache implementation is useful for development and tests, it's recommended to use Caffeine cache in production. You can switch to using Caffeine
the classpath.
forneceadores-service | 2026-10-03T01:01:41.468Z INFO 1 --- [forneceadores-service] [ main] o.s.c.n.eureka.InstanceInfoFactory : Setting initial instance status as: STARTING
forneceadores-service | 2026-10-03T01:01:41.524Z INFO 1 --- [forneceadores-service] [ main] com.netflix.discovery.DiscoveryClient : Initializing Eureka in region us-east-1
forneceadores-service | 2026-10-03T01:01:41.527Z INFO 1 --- [forneceadores-service] [ main] c.n.d.s.r.aws.ConfigClusterResolver : Resolving eureka endpoints via configuration
forneceadores-service | 2026-10-03T01:01:41.548Z INFO 1 --- [forneceadores-service] [ main] com.netflix.discovery.DiscoveryClient : Disable delta property : false
forneceadores-service | 2026-10-03T01:01:41.548Z INFO 1 --- [forneceadores-service] [ main] com.netflix.discovery.DiscoveryClient : Single vip registry refresh property : null
forneceadores-service | 2026-10-03T01:01:41.548Z INFO 1 --- [forneceadores-service] [ main] com.netflix.discovery.DiscoveryClient : Force full registry fetch : false
forneceadores-service | 2026-10-03T01:01:41.549Z INFO 1 --- [forneceadores-service] [ main] com.netflix.discovery.DiscoveryClient : Application is null : false
forneceadores-service | 2026-10-03T01:01:41.549Z INFO 1 --- [forneceadores-service] [ main] com.netflix.discovery.DiscoveryClient : Registered Applications size is zero : true
forneceadores-service | 2026-10-03T01:01:41.549Z INFO 1 --- [forneceadores-service] [ main] com.netflix.discovery.DiscoveryClient : Application version is -1: true
forneceadores-service | 2026-10-03T01:01:41.549Z INFO 1 --- [forneceadores-service] [ main] com.netflix.discovery.DiscoveryClient : Getting all instance registry info from the eureka server
forneceadores-service | 2026-10-03T01:01:41.735Z INFO 1 --- [forneceadores-service] [ main] com.netflix.discovery.DiscoveryClient : The response status is 200
forneceadores-service | 2026-10-03T01:01:41.737Z INFO 1 --- [forneceadores-service] [ main] com.netflix.discovery.DiscoveryClient : Starting heartbeat executor: renew interval is: 30
forneceadores-service | 2026-10-03T01:01:41.739Z INFO 1 --- [forneceadores-service] [ main] c.n.discovery.InstanceInfoReplicator : InstanceInfoReplicator onDemand update allowed rate per min is 4
forneceadores-service | 2026-10-03T01:01:41.742Z INFO 1 --- [forneceadores-service] [ main] com.netflix.discovery.DiscoveryClient : Discovery Client initialized at timestamp 1790989301742 with initial instances cou
nt: 6
forneceadores-service | 2026-10-03T01:01:41.744Z INFO 1 --- [forneceadores-service] [ main] o.s.c.n.e.s.EurekaServiceRegistry : Registering application FORNECEDORES-SERVICE with eureka with status UP
forneceadores-service | 2026-10-03T01:01:41.744Z INFO 1 --- [forneceadores-service] [ main] com.netflix.discovery.DiscoveryClient : Saw local status change event StatusChangeEvent [timestamp=1790989301744, current=
UP, previous=STARTING]
forneceadores-service | 2026-10-03T01:01:41.745Z INFO 1 --- [forneceadores-service] [nfoReplicator-0] com.netflix.discovery.DiscoveryClient : DiscoveryClient_FORNECEDORES-SERVICE/7bdaf6a7415:forneceadores-service:8084: regIs
tering service...
forneceadores-service | 2026-10-03T01:01:41.755Z INFO 1 --- [forneceadores-service] [ main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8084 (http) with context path ''
forneceadores-service | 2026-10-03T01:01:41.756Z INFO 1 --- [forneceadores-service] [ main] .s.c.n.e.s.EurekaAutoServiceRegistration : Updating port to 8084
forneceadores-service | 2026-10-03T01:01:41.764Z INFO 1 --- [forneceadores-service] [ main] c.e.f.ForneceadoresServiceApplication : Started ForneceadoresServiceApplication in 3.742 seconds (process running for 4.088
s)
forneceadores-service | 2026-10-03T01:01:41.776Z INFO 1 --- [forneceadores-service] [nfoReplicator-0] com.netflix.discovery.DiscoveryClient : DiscoveryClient_FORNECEDORES-SERVICE/7bdaf6a7415:forneceadores-service:8084 - regI
station status: 204
forneceadores-service | Hibernate: select count(*) from fornecedor f1_0
forneceadores-service | Hibernate: insert into fornecedor (cnpj,nome,id) values (?,?,default)
forneceadores-service | Hibernate: insert into fornecedor (cnpj,nome,id) values (?,?,default)
forneceadores-service | Hibernate: insert into fornecedor (cnpj,nome,id) values (?,?,default)
forneceadores-service | Hibernate: insert into fornecedor (cnpj,nome,id) values (?,?,default)
forneceadores-service | 2026-10-03T01:01:53.374Z INFO 1 --- [forneceadores-service] [nio-8084-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
forneceadores-service | 2026-10-03T01:01:53.374Z INFO 1 --- [forneceadores-service] [nio-8084-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
forneceadores-service | 2026-10-03T01:01:53.375Z INFO 1 --- [forneceadores-service] [nio-8084-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 1 ms
forneceadores-service | 2026-10-03T01:00:41.550Z INFO 1 --- [forneceadores-service] [trap-executor-0] c.n.d.s.r.aws.ConfigClusterResolver : Resolving eureka endpoints via configuration
forneceadores-service | 2026-10-03T01:11:41.551Z INFO 1 --- [forneceadores-service] [trap-executor-0] c.n.d.s.r.aws.ConfigClusterResolver : Resolving eureka endpoints via configuration
forneceadores-service | 2026-10-03T01:16:41.551Z INFO 1 --- [forneceadores-service] [trap-executor-0] c.n.d.s.r.aws.ConfigClusterResolver : Resolving eureka endpoints via configuration

guilherme@guilherme-china: ~/api-vendas-AT$ java -Datividade-guilherme_china [?] [ (us-east-1)]
^C
guilherme@guilherme-china$
```

Figura 3 — Inicialização do novo microserviço.

Exercício 4 — Cinco fornecedores no H2

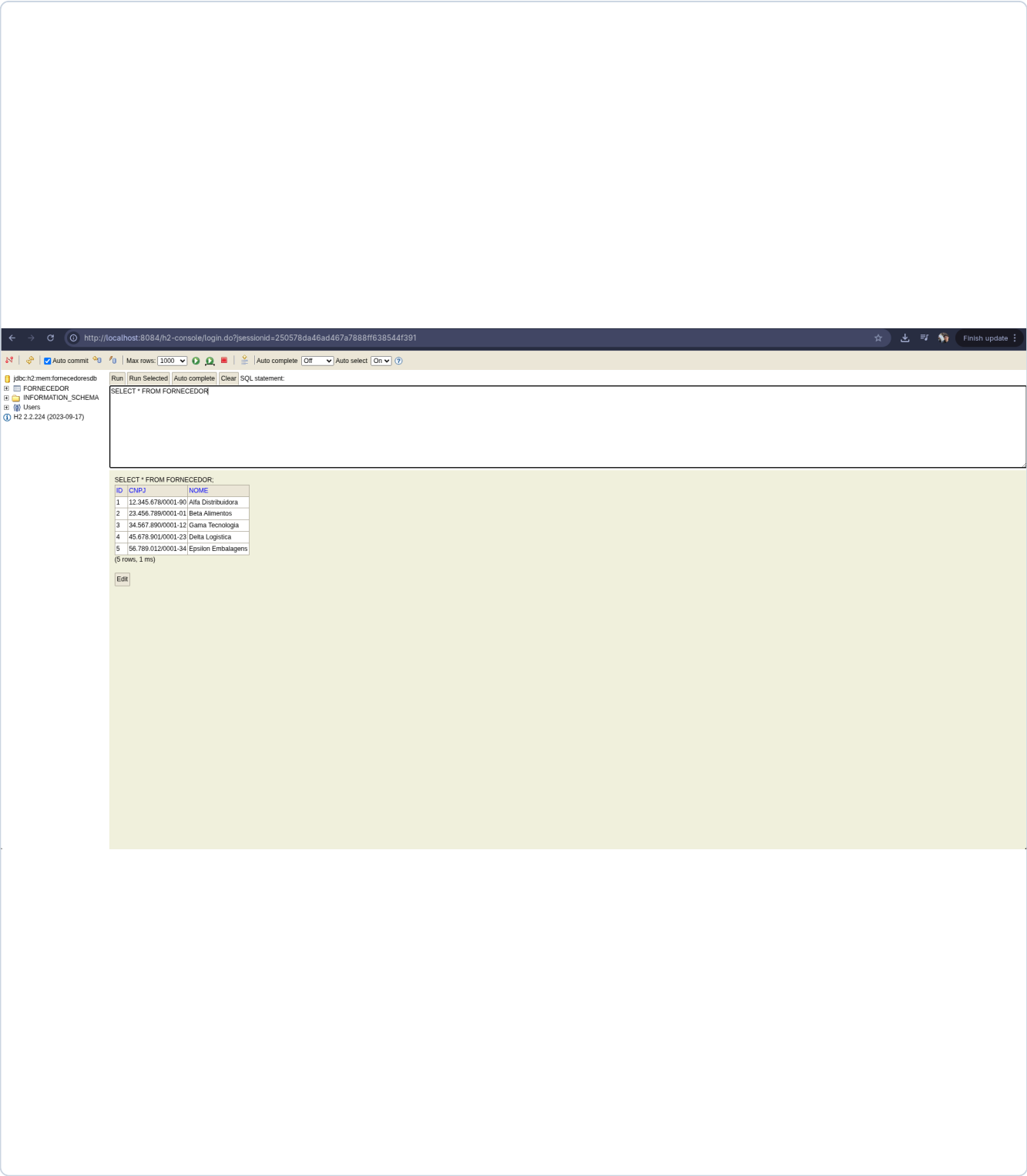


Figura 4 — Dados carregados automaticamente no H2.

Exercício 5 — Endpoints GET e status 404

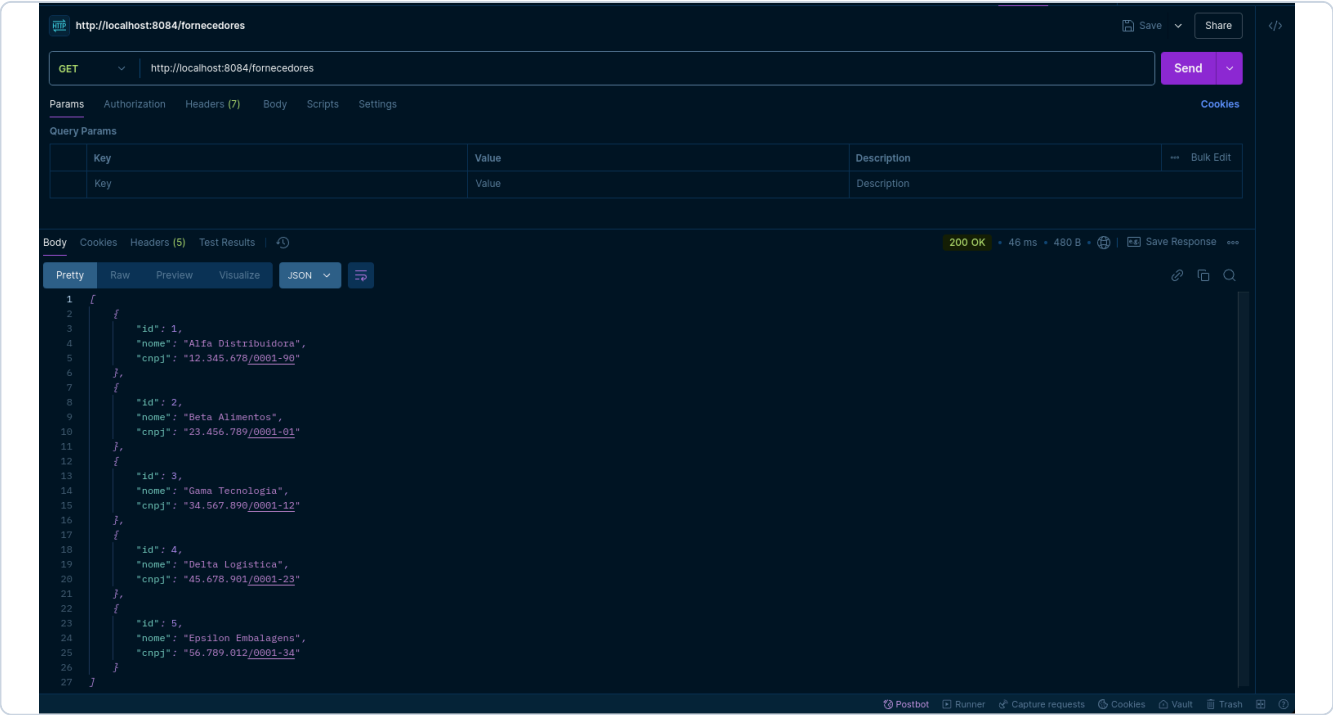


Figura 5A — Listagem de fornecedores com status 200 OK.

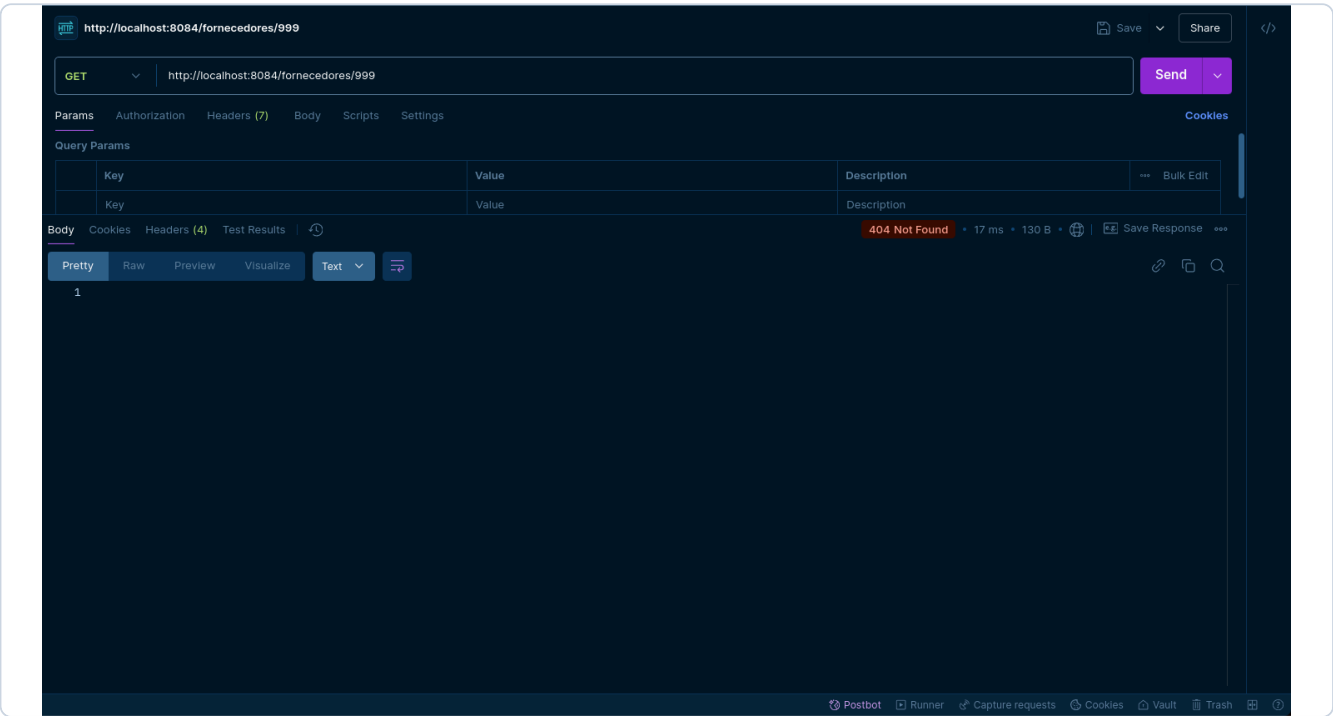


Figura 5B — Consulta de ID inexistente com status 404 Not Found.

Exercício 6 — fornecedores-service no Eureka

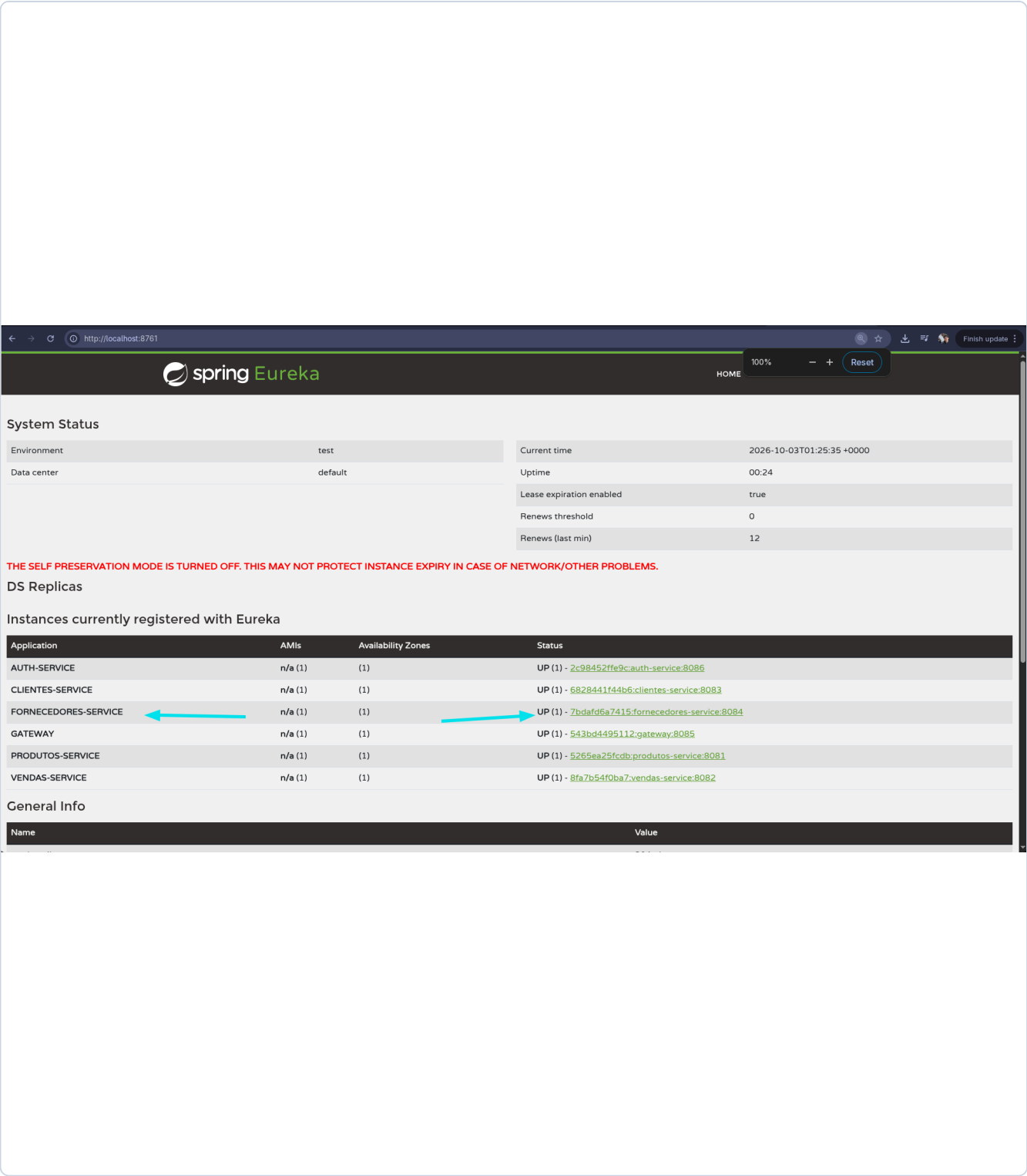


Figura 6 — Registro do fornecedores-service no Eureka.

Exercício 7 — Configuração centralizada

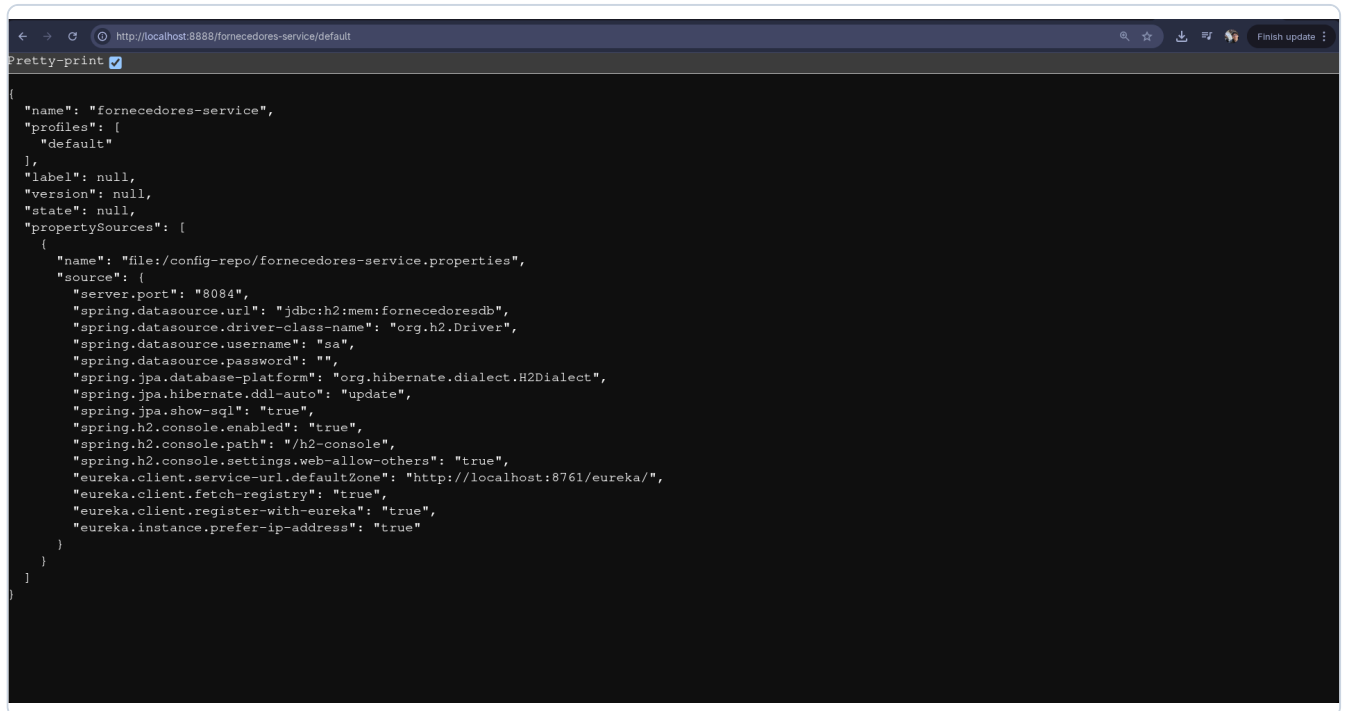


Figura 7A — Propriedades do serviço fornecidas pelo Config Server.

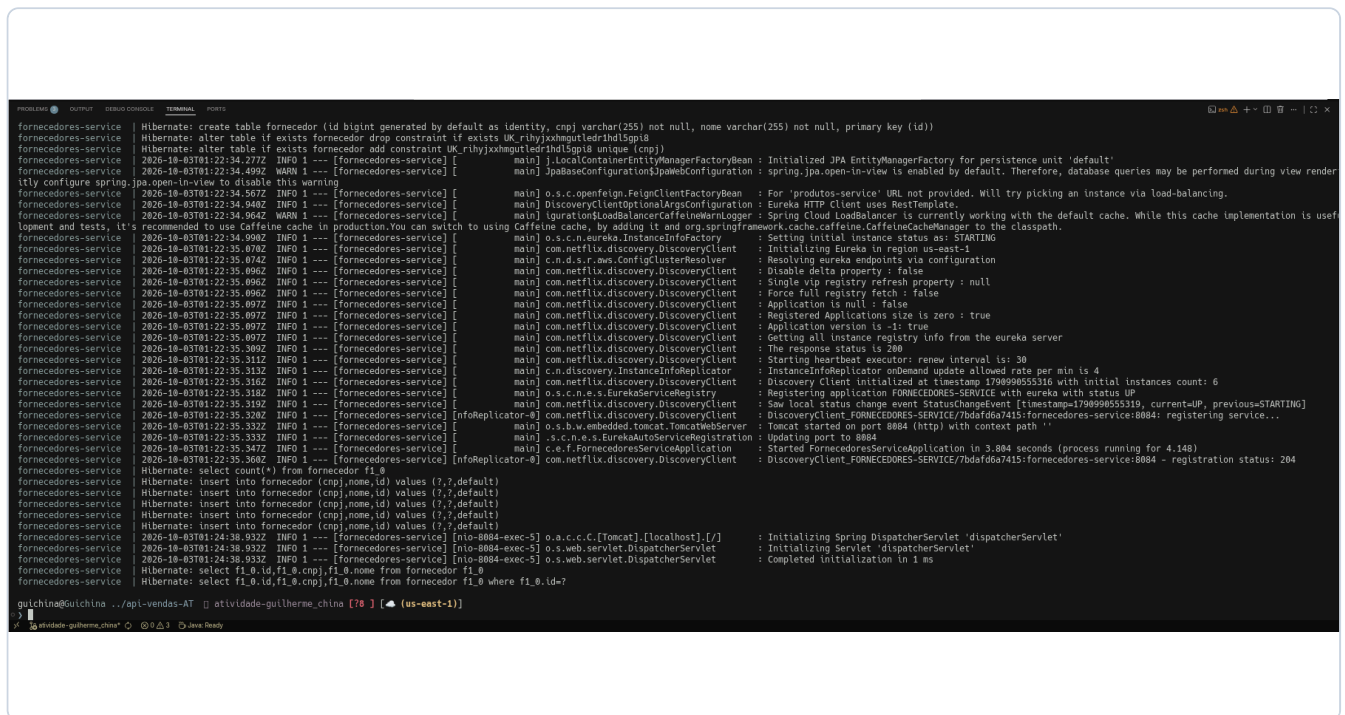


Figura 7B — Serviço iniciado na porta 8084 configurada externamente.

Exercício 8 — Acesso pelo Gateway

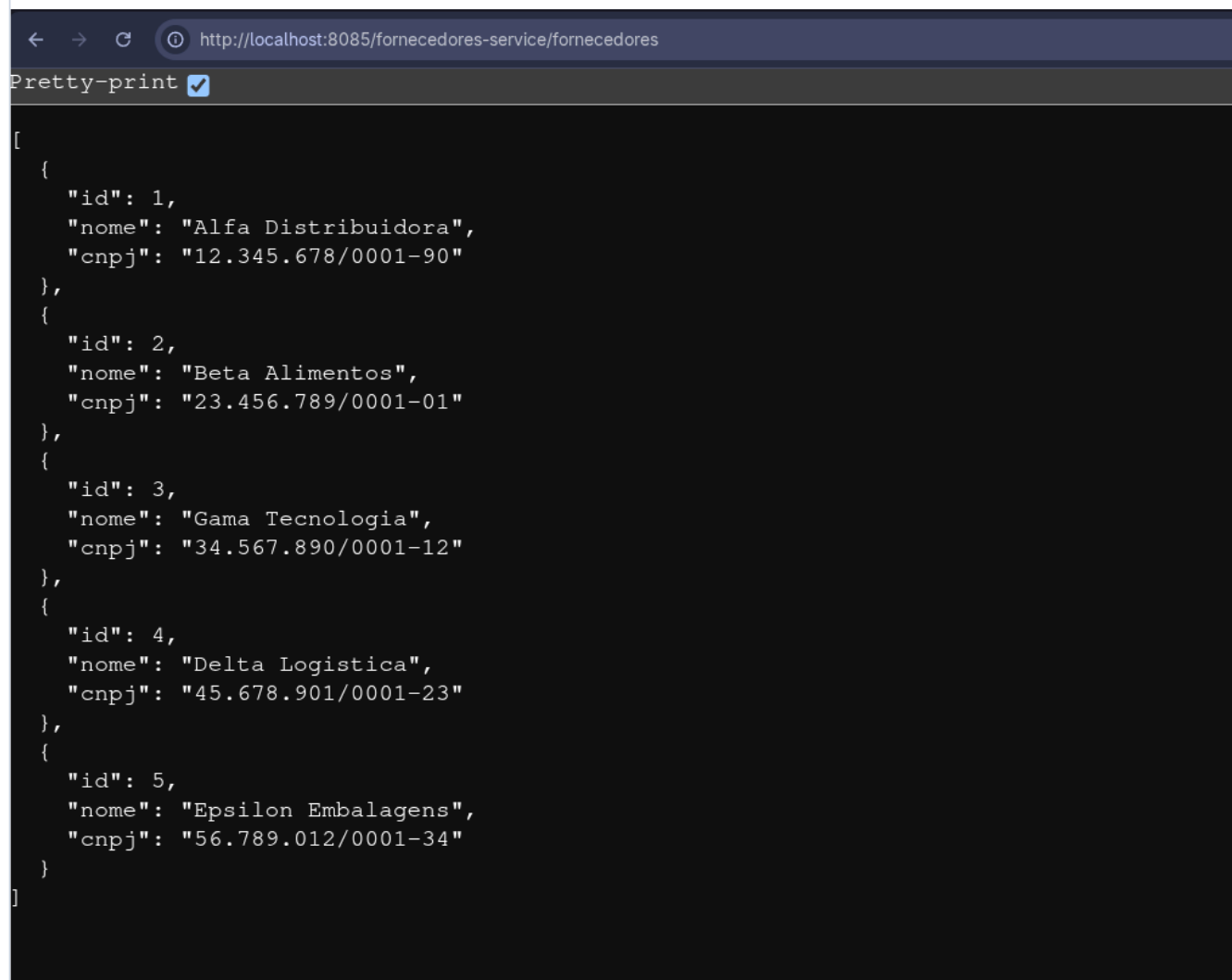


Figura 8 — Roteamento dinâmico pelo Spring Cloud Gateway.

Exercício 9 — Cadastro com POST

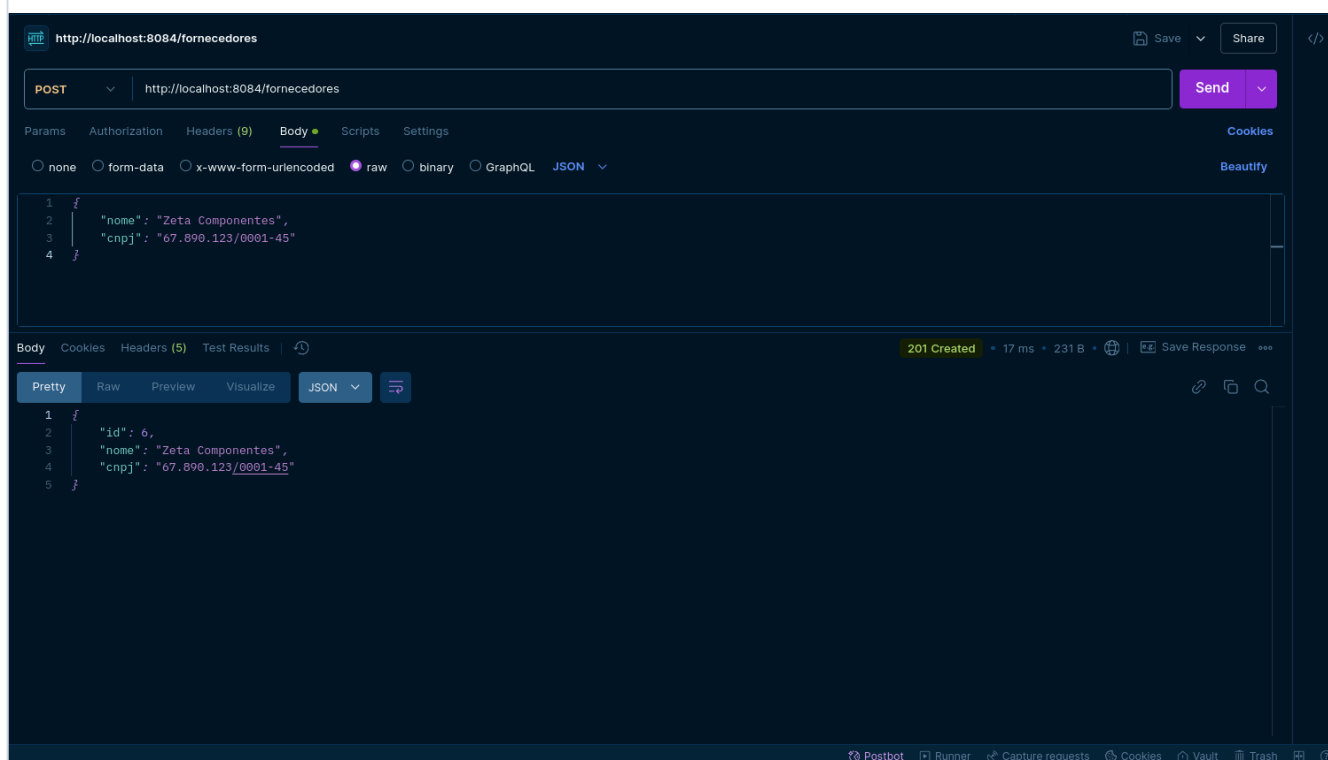


Figura 9 — Novo fornecedor persistido.

Exercício 10 — Consulta ao produtos-service com Feign

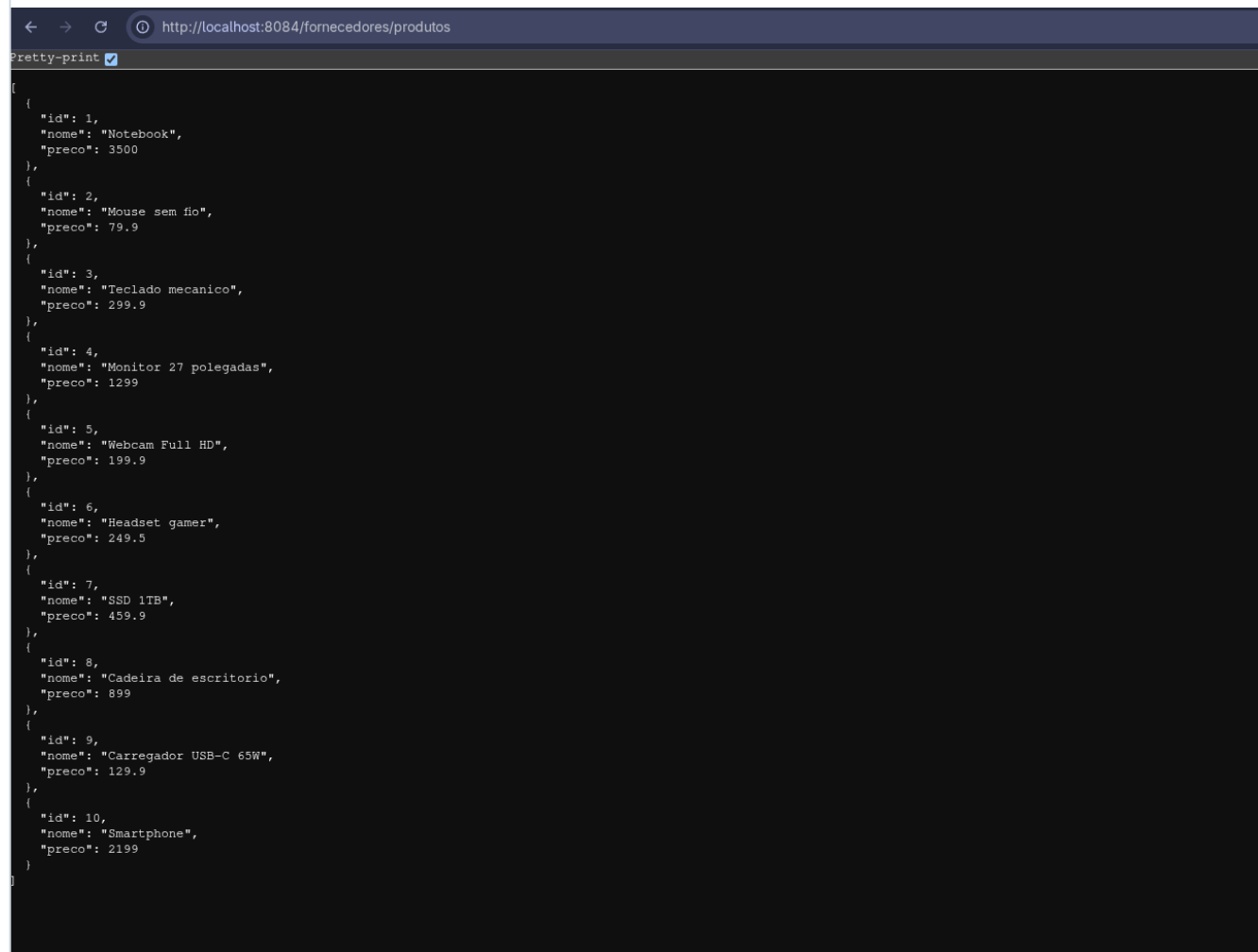


Figura 10 — Comunicação entre microserviços via OpenFeign e Eureka.

Exercício 11 — Docker Compose e Gateway

```
guichina@guichina: ~/api-vendas-AT [?] [us-east-1]
$ docker compose ps
NAME                IMAGE                                COMMAND                                SERVICE    CREATED   STATUS    PORTS
auth-service        api-vendas-at-auth-service         "java -jar app.jar"                auth-service    29 minutes ago    Up 29 minutes    0.0.0.0:8086->8086/tcp, [::]:8086->8086/tcp
clientes-service    api-vendas-at-clientes-service     "java -jar app.jar"                clientes-service 29 minutes ago    Up 29 minutes    0.0.0.0:8083->8083/tcp, [::]:8083->8083/tcp
config-server       api-vendas-at-config-server        "java -jar app.jar"                config-server    29 minutes ago    Up 29 minutes (healthy)    0.0.0.0:8888->8888/tcp, [::]:8888->8888/tcp
eureka-server       api-vendas-at-eureka-server        "java -jar app.jar"                eureka-server    29 minutes ago    Up 29 minutes (healthy)    0.0.0.0:8761->8761/tcp, [::]:8761->8761/tcp
fornecedores-service api-vendas-at-fornecedores-service "java -jar app.jar"                fornecedores-service 28 minutes ago    Up 7 minutes    0.0.0.0:8084->8084/tcp, [::]:8084->8084/tcp
gateway             api-vendas-at-gateway              "java -jar app.jar"                gateway          32 minutes ago    Up 31 minutes    0.0.0.0:8085->8085/tcp, [::]:8085->8085/tcp
produtos-service    api-vendas-at-produtos-service     "java -jar app.jar"                produtos-service 29 minutes ago    Up 29 minutes (healthy)    0.0.0.0:8081->8081/tcp, [::]:8081->8081/tcp
vendas-service      api-vendas-at-vendas-service       "java -jar app.jar"                vendas-service    29 minutes ago    Up 29 minutes    0.0.0.0:8082->8082/tcp, [::]:8082->8082/tcp

guichina@guichina: ~/api-vendas-AT [?] [us-east-1]
$
```

Figura 11A — Infraestrutura iniciada pelo Docker Compose.

```
← → ↻ http://localhost:8085/fornecedores-service/fornecedores
Pretty-print ☒
[
  {
    "id": 1,
    "nome": "Alfa Distribuidora",
    "cnpj": "12.345.678/0001-90"
  },
  {
    "id": 2,
    "nome": "Beta Alimentos",
    "cnpj": "23.456.789/0001-01"
  },
  {
    "id": 3,
    "nome": "Gama Tecnologia",
    "cnpj": "34.567.890/0001-12"
  },
  {
    "id": 4,
    "nome": "Delta Logistica",
    "cnpj": "45.678.901/0001-23"
  },
  {
    "id": 5,
    "nome": "Epsilon Embalagens",
    "cnpj": "56.789.012/0001-34"
  },
  {
    "id": 6,
    "nome": "Zeta Componentes",
    "cnpj": "67.890.123/0001-45"
  }
]
```

Figura 11B — Resposta integrada pelo Gateway no ambiente Docker.

Exercício 12 — GitHub Actions

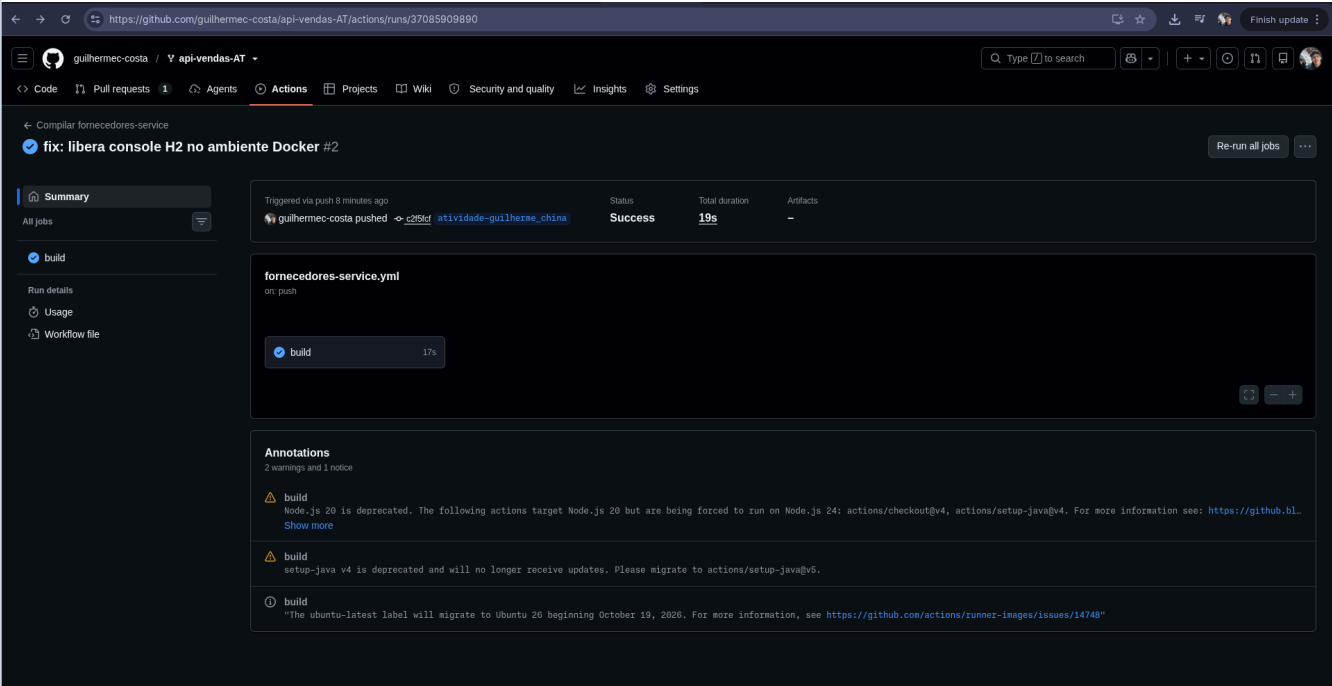


Figura 12 — Compilação Maven automatizada com Java 17.